

Orthogonal trajectories.

Step 1: Find $\frac{dy}{dx}$ of c , put value of c in $\frac{dy}{dx}$

Step 2: If eqn can be solved with no V , just do it!

Ex: 1, 2, 3, 4 | Ex: 5: needs $V = \frac{y}{x}$

Step 3: solve eqn.

1) $y^2 = cx$ 2) $cx^2 + y^2 = 1$ 3) $y = \frac{cx^2}{x+1}$ ~~4) $y = e^x$~~ 4) $y = e^{cx}$

5) $x^2 + y^2 = cx^3$

(2) Given,

$$ex^2 + y^2 = 1$$

$$\Rightarrow y^2 = 1 - ex^2$$

$$\Rightarrow 2y \frac{dy}{dx} = -2xe$$

$$\Rightarrow y \frac{dy}{dx} = -x \cdot \frac{1-y^2}{x^2} \quad \left[\because e = \frac{1-y^2}{x^2} \right]$$

$$\Rightarrow y \frac{dy}{dx} = \frac{y^2 - 1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 - 1}{xy}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-xy}{y^2 - 1}$$

$$\Rightarrow (y^2 - 1) dy = (-xy) dx$$

$$\Rightarrow \left(\frac{y^2 - 1}{y} \right) dy = -x dx$$

$$\Rightarrow \int \frac{y^2}{y} dy - \int \frac{1}{y} dy = - \int x dx$$

$$\Rightarrow \frac{y^2}{2} - \ln y = -\frac{x^2}{2}$$

$$\Rightarrow \ln y - \frac{y^2}{2} = \frac{x^2}{2} + c$$

$$\Rightarrow \ln y - \frac{y^2}{2} - \frac{x^2}{2} = c$$

(Ans.)

(3) Given,

$$y = \frac{e^{x^2}}{x+1}$$

$$\Rightarrow \frac{dy}{dx} = e \cdot \frac{d}{dx} \left(\frac{x^2}{x+1} \right)$$

$$\Rightarrow \frac{dy}{dx} = e \cdot \frac{(x+1) \cdot 2x - x^2 \cdot 1}{(x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(x+1)}{x^2} \times \frac{(x+1) \cdot 2x - x^2}{(x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(2x^2 + 2x) - x^2 y}{x^2(x+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x(2xy + 2y - x^2 y)}{x^2(x+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(2x+2-x^2)}{x(x+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(x+2)}{x(x+1)}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal

$$\therefore \frac{dy}{dx} = \frac{-x(x+1)}{y(x+2)}$$

$$\Rightarrow y(x+2) dy = -x(x+1) dx$$

$$\Rightarrow y dy = \left(\frac{-x^2 - x}{x+2} \right) dx$$

$$\Rightarrow \int y \, dy = \int \frac{-x^2 - x + 2 - 2}{x+2} \, dx$$

$$\Rightarrow \int y \, dy = \int \frac{-x^2 - 2x + x + 2 - 2}{x+2} \, dx$$

$$\Rightarrow \int y \, dy = \int \frac{-x(x+2) + 1(x+2) - 2}{(x+2)} \, dx$$

$$\Rightarrow \int y \, dy = \int \left\{ \frac{(x+2)(1-x)}{x+2} - \frac{2}{x+2} \right\} \, dx$$

$$\Rightarrow \int y \, dy = \int \frac{(x+2)(1-x)}{x+2} \, dx - 2 \int \frac{1}{x+2} \, dx$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} - 2 \ln(x+2) + c$$

$$\Rightarrow \frac{y^2}{2} + \frac{x^2}{2} + 2 \ln(x+2) = c$$

(Ans.)

5) Given,

$$x^2 + y^2 = cx^3 \quad \text{--- (1)}$$

$$\therefore 2x + 2y \frac{dy}{dx} = 3cx^2$$

$$\Rightarrow 2y \frac{dy}{dx} = 3cx^2 - 2x$$

$$\Rightarrow 2y \frac{dy}{dx} = 3 \left(\frac{x^2 + y^2}{x^3} \right) \cdot x^2 - 2x$$

$$\Rightarrow 2y \frac{dy}{dx} = \frac{3x^2 + 3y^2}{x} - 2x$$

$$\Rightarrow 2y \frac{dy}{dx} = \frac{3x^2 + 3y^2 - 2x^2}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2} \quad \text{--- (2)}$$

Let,

$$v = \frac{y}{x}$$

$$\Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

put $\frac{dy}{dx}$ and y in (2),

$$v + x \frac{dv}{dx} = \frac{-2x \cdot vx}{x^2 + 3(vx)^2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-2vx^2}{x^2 + 3v^2x^2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-2v}{1 + 3v^2} = -v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-2v - v - 3v^3}{1 + 3v^2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-3v^3 - 3v}{1+3v^2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-3(v^3+v)}{1+3v^2}$$

$$\Rightarrow \frac{1+3v^2}{v^3+v} dv = \frac{-3}{x} dx$$

$$\Rightarrow \int \frac{1+3v^2}{v^3+v} dv + \int \frac{3}{x} dx = 0$$

$$\Rightarrow \ln(v^3+v) + 3\ln x = \ln c$$

$$\Rightarrow \ln(v^3+v) + \ln x^3 = \ln c$$

$$\Rightarrow \ln \{x^3 \cdot (v^3+v)\} = \ln c$$

$$\Rightarrow x^3 (v^3+v) = c$$

$$\Rightarrow x^3 \left(v \frac{y^3}{x^3} + \frac{y}{x} \right) = c$$

$$\Rightarrow x^3 \times \frac{y^3}{x^3} + x^3 \frac{y}{x} = c$$

$$\Rightarrow y^3 + x^2 y = c$$

(Ans.)

4) Given,
 $y = e^{cx}$

$$\Rightarrow \ln y = cx$$

$$\Rightarrow c = \frac{\ln y}{x}$$

$$\therefore y = e^{cx}$$

$$\Rightarrow \frac{dy}{dx} = ce^{cx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\ln y}{x} \cdot y \quad \left[\because c = \frac{\ln y}{x} ; y = e^{cx} \right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \ln y}{x}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-x}{y \ln y}$$

$$\Rightarrow y \ln y \, dy = -x \, dx$$

$$\Rightarrow \int y \ln y \, dy + \int x \, dx = 0$$

$$\Rightarrow \ln y \int y \, dy - \int \left[\frac{d}{dy} (\ln y) \int y \, dy \right] dy + \frac{x^2}{2} = 0$$

$$\Rightarrow \ln y \cdot \frac{y^2}{2} - \int \left(\frac{1}{y} \cdot \frac{y^2}{2} \right) dy + \frac{x^2}{2} = c$$

$$\Rightarrow \ln y \cdot \frac{y^2}{2} - \frac{1}{2} \times \frac{y^2}{2} + \frac{x^2}{2} = c$$

$$\Rightarrow \frac{y^2 \ln y}{2} - \frac{y^2}{4} + \frac{x^2}{2} = c$$

(Ans)

“TATTA”

Given,

$$y^2 = ex$$

$$\therefore 2y \frac{dy}{dx} = e$$

$$\Rightarrow 2y \frac{dy}{dx} = \frac{y^2}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2}{2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{2x}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-2x}{y}$$

$$\Rightarrow y dy = -2x dx$$

$$\Rightarrow \int y dy = -2 \int x dx$$

$$\Rightarrow \frac{y^2}{2} = -2 \cdot \frac{x^2}{2} + e$$

$$\Rightarrow \frac{y^2}{2} = -x^2 + e$$

$$\Rightarrow y^2 = -2x^2 + e$$

$$\Rightarrow y^2 + 2x^2 = e$$

(Ans.)